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Discussion Agenda 08/17/2014

Below is a discussion agenda / live progress report in an approximate order of importance for the brainstorming session of the 3rd AGORA Workshop @ University of California Santa Cruz (Aug. 15–17, 2014; see the [program](#))

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Check out the important items in red color below!

(1st revision: 08/20/2014)

[1. Get Yourself Up-To-Date!](#)

- (1) Did you miss this year's Workshop? Make sure to get yourself up-to-date with the first talk of the 3rd AGORA Workshop, "**Status of the AGORA Project: Two Years After Conception**" by Joel Primack and Ji-hoon Kim (Aug. 15; [slides](#)).
- (2) Also check out Oscar Agertz's talk, "**To-do Items for Papers 3 and 4**" (Aug. 16; [slides](#), [SeeVogh screenshot](#)), and two important [points](#) Joel Primack tried to make at the beginning of the working sessions.
- (3) If you want to plan on using NERSC resources for AGORA, you may not want to miss Miguel Rocha's talk, "**yt+AGORA Pipeline on NERSC**" (Aug. 17; [slides](#)).

[2. Progress Report \(as of August 2014\)](#)

We are actively making a lot of progress, chiefly, in three different fronts below (numbered tentatively).

	PAPER "2"	PAPER "3"	PAPER "4"																						
GOALS	Run Proof-of-concept-DM (~1e11 Ms halo)	- Implement and test common astrophysics package - Run Proof-of-concept-NOSUBGRID (~1e11 Ms halo)	Calibrate SF/Fbck recipes using common disk ICs																						
LEADERS	Oliver Hahn (ETH)	- Nick Gnedin (Fermilab) - Piero Madau (UCSC) - Britton Smith (Edinburgh)	- Oscar Agertz (Surrey) - Romain Teyssier (Zurich) - James Wadsley (McMaster)																						
	- Working Groups engaged: - III (common cosmo ICs) - IV (common analysis) - VI (dwarf galaxies) - Code representatives: (discussion participants)	- Working Groups engaged: - I (common physics) - III (common cosmo ICs) - Code representatives: (Grackle implementers)	- Working Groups engaged: - II (common disk ICs) - IV (common analysis) - V (subgrid physics) - Code representatives: (discussion participants)																						
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PARTICIPANTS	CHANGA	(Quinn)		Letiner	ART-II	Feldmann
	ENZO	Kim, Simpson, Wise	CHANGA	(Quinn)	CHANGA	Quinn
	GADGET	Choi, Hahn, Hobbs, Nagamine, Onorbe, Rocha, Todoroki	ENZO	Hummels, Kim, Simpson, Smith, Turk, Wise	ENZO	Butler, Forbes, Hummels, Kim
			GADGET	Choi, Nagamine,	GADGET	Hobbs, Nagamine, Thompson, Todoroki
	GASOLINE	Shen, Zolotov	GASOLINE	Madau, Mayer, Shen, Wadsley	GASOLINE	Shen, Wadsley, Benincasa
	GCD+	Kawata	GEAR	Revaz	GEAR	Revaz
	PKDGRAV2	Guedes, (TBA)	RAMSES	Agertz, Teyssier	RAMSES	Agertz, Butler, DeGraf, Teyssier, Nickerson
	RAMSES	Hahn, Teyssier				
DELIVERABLES IN 2014	• Push out PAPER "2" in the fall of 2014		• Push out PAPER "3" in 2014		• Push out PAPER "4" in the winter of 2014	
PROGRESS MADE OR BEING MADE	• Please visit Paper 2's Workspace where Oliver Hahn has been updating the progress report.		• A telecon was held among Paper "3" participants on June 13, 2014. Please take a look at the discussion result and many ongoing actions afterwards in this document coordinated by Nick Gnedin and Britton Smith. • GRACKLE or Nick Gnedin's dummy API for GRACKLE is now implemented in all codes. Nick and others will now code up an actual GRACKLE wrapper which will then be tested in isolated disk runs (Paper "4") or in a $10^{11} M_{\text{sun}}$ cosmological halo. • Thanks to the great effort by Britton Smith, functioning and tested GRACKLE APIs for Fortran, C, and C++ codes are now available. See his announcement and the document linked above for more information.		• Please visit Paper 4's Workspace where Oscar Agertz has been updating the progress report. See also his talk above. • yt-3.0 is released on Aug. 4, 2014 thanks in part to many AGORA members including Matthew Turk. See the announcement. • yt+AGORA pipeline is being built on the NERSC system by Miguel Rocha, Alex Bogert, Matthew Turk, Joel Primack, and many others. See their talks above, or this section below for more information.	
TIMELINE IN 2014	• Please visit Paper 2's Workspace.		• (to be filled)		• Please visit Paper 4's Workspace.	
OTHER LINKS	• Please see the progress report in Paper 2's Workspace and/or in WG III's Workspace. • MUSIC Google group		• GRACKLE Google group • GRACKLE download & documentation		• Please see the progress report in Paper 4's Workspace and/or in WG V's Workspace. • yt-AGORA Google group	

3. SMBH Accretion and Feedback Recipes

The following is a SMBH accretion and feedback recipes proposal based on a Working Group XIII discussion led by Alex Hobbs on Aug. 16, 2014. (Discussion participants: DeGraf, Durier, Hobbs, Kim, Primack) The WG XIII will very soon vote on this proposal after more discussion, especially regarding item #(4) below. This proposal does *not* preclude you from running AGORA simulations with your own favorite recipes.

(1) It was decided to restrict the SMBH runs to only the cosmological simulations, and drop the plan to re-run the isolated discs with a central black hole. For the cosmological SMBH runs then the proposal is the following:

(2) **Accretion model:** While there are a number of very promising recent models to choose from, for the purposes of a baseline comparison (and to ensure that the model we use has been tested in both SPH and grid-based codes) we will keep it simple and adopt one of the standard Bondi-Hoyle variants from the literature. We therefore ask all working group members to vote on either:

- (i) Standard Bondi-Hoyle
- (ii) Bondi-Hoyle with alpha modification based on ratio of n_H (Booth & Schaye 2009)

(3) **Feedback model:** Again we need a simple model that has been borne out in the literature across both types of code and for this reason we will use standard thermal energy injection to couple the feedback to the accretion rate. For the precise implementation of this we ask all working group members to vote on either:

- (i) Simple kernel-weighted feedback, energy is released every timestep according to accretion rate
- (ii) 'Bottled-up' feedback where energy is released once above a certain threshold (e.g. Booth & Schaye 2009)

In both cases the energy is injected over the neighbours/nearby grid cells weighted by a kernel function.

(4) **Seeding, Merging, etc:** We will have a further web conference at a later date to discuss possible BH seeding approaches and decide on one that can be implemented easily in both SPH and grid-based codes.

4. NERSC Resources: Account Request, Recaps

The following is about the new NERSC resources for AGORA ("Data Pilot" Program, PI: Joel Primack), and how you can get an account on it. Find out more about this new AGORA mass storage in Primack+Kim's [talk](#) above.

(1) Organize the storage by code groups ("workspace"). Initially one user@NERSC per each code group. Different versions of the same code will be considered as different code groups (e.g. Gadget-3-UNLV, Gadget-3-SPHS, etc). We will soon build a new page here on the Google Project Workspace on how to use the NERSC resources.

(2) NERSC has installed one Kepler K20 and one Kepler K40 card in the system for AGORA analysis and visualization. It plans to purchase more of the latest Nvidia graphics cards with the Kepler architecture, with matching funds up to \$10,000 from UC-HiPACC.

(3) yt+AGORA pipeline is being built on the NERSC system by Miguel Rocha, Alex Bogert, Matthew Turk, Joel Primack, and many others. Please see their [talks](#) above for more information. The prototype of the Science Gateway of the AGORA Project is [here](#). If you represent a code group but don't have a NERSC user account, visit [this page](#). Select "standard" account and find the repository name "agora". Joel Primack will add your name to the group when approved.

[5. Strategies to Move the Project Forward](#)

- (1) Telecons every two weeks for each paper? Maybe not. But let's set up solid timelines for each paper to motivate participants.
- (2) Mini-workshop for individual paper groups to write papers: UC-HIPACC@UCSC and KIPAC@Stanford are willing to provide funding opportunities for working group leaders to organize mini-workshops (~10 people). An example was the [yt-AGORA mini-workshop](#) at UCSC in Mar. 2014.

[6. Recaps](#)

(1) **Strategy to Store Simulation Outputs** (based on the proposal by Peter Behroozi and Joel Primack; **NOTE that this is a default policy -- strategies may vary per individual project**)

- Each of the participating codes will generate large quantities of unprocessed, intermediate data, in the form of "checkpoints" describing the state of the simulation at a given time. We recommend total 201 timesteps: the first 21 timesteps equally spaced in redshift, $\Delta z = 0.5$, from $z=20$ to $z=10$, and the remaining 180 timesteps equally spaced in $\log(\text{expansion parameter})$, $\Delta \log(a) = |\log(1/11)/180|$, from $z=10$ to $z=0$. Plus redshift snapshots at $z = 6.0, 3.0, 2.0, 1.0, 0.5, 0.2, 0.0$ at the very least. The explicit list of outputs can be downloaded [here](#) (thanks to Junhwan Choi).
- Each group is advised to make sure that they can restart the simulation from a point of your interest by storing the "checkpoints", and also that their outputs are "analyzable." (Because "checkpoints" might not necessarily be "analyzable" in some codes.) This way, one may add more timesteps in the time period of your interest.
- Each group may be advised to store more timesteps depending on the science questions being pursued (e.g. ~1000 timesteps to study the orbital motions of satellite galaxies). Each group may also want to store additional outputs at slightly earlier redshift, shifted by $\Delta z \sim 0.05$. This extra information will help us investigate whether an inter-platform offset in time-stepping causes timing differences for halo mergers, for example.

(2) **Useful Links:**

- Collection of softwares relevant to AGORA effort: [here](#)
- Quick links to the important pages in AGORA Workspace: [here](#)
- Internal policy regarding the disclosure of AGORA simulation results: [here](#)